

TAUROVO, Russian Federation

WMO: 280970

Lat: 59.5855N

Lon: 73.2891E

Elev: 198

StdP: 14.59

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.6	(c) -32.8	(d) -43.3	(e) 0.5	(f) -37.9	(g) -37.6	(h) 0.6	(i) -32.1	(j) 9.1	(k) 14.6	(l) 7.7	(m) 12.7	(n) 1.0	(o) 150	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 22.4	(c) 83.9	(d) 66.4	(e) 81.0	(f) 65.7	(g) 77.5	(h) 64.2	(i) 69.0	(j) 79.3	(k) 67.4	(l) 77.3	(m) 65.7	(n) 74.5	(o) 4.0	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	95.6	73.9	63.8	89.3	71.0	62.2	84.4	69.6	33.3	79.5	32.0	77.9	30.7	74.7	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.6	(b) 8.3	(c) 7.5	(e) -47.0	(f) 88.1	(g) 5.8	(h) 1.6	(i) -51.2	(j) 89.3	(k) -54.6	(l) 90.2	(m) -57.8	(n) 91.1	(o) -62.0	(p) 92.3		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.9	-3.7	3.8	18.3	34.4	44.3	59.4	63.4	57.5	45.6	31.3	11.8	3.2
	DBStd	25.59	15.95	15.85	12.74	8.49	9.20	8.68	6.66	7.03	7.59	9.33	15.66	16.51
	HDD50	8034	1664	1293	983	469	231	23	1	14	176	581	1147	1452
	HDD65	12552	2129	1713	1448	917	642	205	109	249	582	1044	1597	1917
	CDD50	1070	0	0	0	2	55	306	416	246	44	1	0	0
	CDD65	115	0	0	0	0	1	39	59	16	0	0	0	0
	CDH74	1548	0	0	0	0	79	554	702	206	7	0	0	0
	CDH80	364	0	0	0	0	13	144	176	31	0	0	0	0
(14) Wind	WSAvg	3.6	3.1	3.3	4.1	4.2	4.6	3.6	3.5	3.2	3.4	3.8	3.5	3.2
(15) Precipitation	PrecAvg	27.8	1.8	1.4	1.9	1.7	2.2	2.8	2.8	3.2	2.3	2.7	2.8	2.3
	PrecMax	36.1	2.8	2.4	4.3	2.6	4.7	4.6	5.6	5.4	3.8	4.9	4.4	4.4
	PrecMin	22.2	0.3	0.2	0.3	0.3	0.9	1.5	0.9	1.5	0.7	1.1	1.1	0.9
	PrecStd	3.1	0.7	0.5	0.9	0.6	0.7	0.8	1.1	0.9	0.8	0.9	0.8	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.3	38.0	44.6	66.6	82.2	87.7	89.9	83.6	74.1	61.0	39.4	32.7
		MCWB	28.3	34.4	35.2	54.3	60.1	67.6	71.7	68.2	59.7	51.8	37.9	31.6
	2%	DB	25.5	33.3	40.2	59.0	75.1	84.3	84.1	79.3	67.3	52.2	35.7	30.4
		MCWB	24.5	31.5	33.5	47.3	57.6	65.9	67.3	66.3	57.8	47.2	34.0	29.3
	5%	DB	22.4	30.6	37.1	54.3	69.5	80.8	81.3	75.5	62.5	47.9	33.3	27.1
		MCWB	21.4	29.0	32.3	44.7	54.5	65.5	66.1	64.2	55.0	44.7	32.2	26.1
	10%	DB	19.0	25.2	34.6	49.8	62.7	76.7	77.8	71.9	58.6	44.5	31.4	24.3
		MCWB	18.1	23.6	31.2	41.9	51.7	63.3	65.1	62.4	53.4	41.8	30.5	23.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.6	34.4	36.7	54.3	61.9	70.3	73.5	69.8	62.0	52.1	38.1	32.2
		MCDB	29.5	36.4	40.6	65.5	79.6	82.0	88.5	78.5	71.6	58.6	39.4	32.6
	2%	WB	24.8	31.9	34.8	49.3	58.4	68.1	69.9	67.6	58.3	47.8	34.5	29.4
		MCDB	25.6	32.9	38.5	56.9	71.6	80.5	79.9	76.5	64.5	50.7	35.3	30.3
	5%	WB	21.5	29.2	33.1	45.6	55.8	66.5	67.9	65.7	56.2	45.2	32.4	26.4
		MCDB	22.3	30.8	36.0	53.0	67.8	77.6	77.4	73.3	61.5	48.0	33.1	27.2
	10%	WB	18.3	23.6	31.5	42.0	52.6	64.4	66.3	63.7	53.4	42.2	30.6	23.5
		MCDB	19.1	25.1	34.5	48.8	62.4	74.7	75.1	70.0	58.2	44.4	31.4	24.3
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	18.8	21.0	20.0	21.2	22.2	22.4	21.5	17.3	11.8	14.3	13.3
		MCDBR	13.8	13.0	19.7	27.8	33.5	28.9	27.4	27.3	24.3	17.7	11.6	10.6
		MCWBR	13.6	11.6	14.5	17.5	17.6	13.0	12.3	14.7	14.5	12.9	10.9	10.2
	5% WB	MCDBR	13.5	12.8	16.2	25.5	29.4	25.1	24.3	24.4	22.6	15.8	11.2	10.1
		MCWBR	13.2	11.9	12.6	16.7	16.8	12.8	12.2	13.6	14.2	12.2	10.8	9.7
(40) Clear-Sky Solar Irradiance	taub		0.269	0.309	0.364	0.387	0.394	0.405	0.422	0.415	0.365	0.335	0.287	0.259
	taud		1.994	2.009	1.938	2.058	2.279	2.325	2.271	2.266	2.361	2.275	2.069	1.938
	Ebn at Noon		152	209	232	254	264	263	254	245	240	205	143	105
	Edh at Noon		16	28	41	43	38	36	38	35	27	22	15	11
(44) All-Sky Solar Radiation	RadAvg		162	456	908	1347	1536	1706	1634	1186	744	346	148	86
	RadStd		15	38	71	138	143	157	179	107	82	44	32	10

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-662	N/A	N/A

Nomenclature: See separate page